

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Module - II

(Backend Development Using Node.js, Express, and MongoDB)

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
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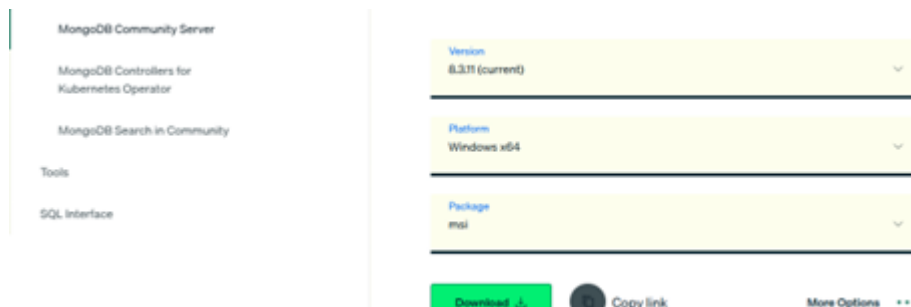
AIM :- MongoDB Basics

A) Install and start MongoDB.

STEP 1 :- Go to the official MongoDB website and download **MongoDB Community Server for Windows**.

Choose:

- **Version:** Current/available version
- **Platform:** Windows
- **Package:** MSI



STEP 2 :- Download the **.msi installer**. After installation, **create a database**.

Click **ADD DATA** → **Insert Document**.

Create Database

Database Name

CollegeDB

Collection Name

Students

☐ Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences

(e.g. Custom collation, Clustered collections)

Cancel

Create Database

B) Create a database and collection

STEP 1 :- Click the + (plus) symbol beside **localhost** to create a database.

Enter:

- **Database Name:** CollegeDB
- **Collection Name:** Students

Insert Document

To collection CollegeDB.Students

```
1 {  
2   "rollNo": 181,  
3   "name": "Saqlain",  
4   "course": "B.Sc. IT",  
5   "marks": 90  
6 }
```

```
> use("CollegeDB")  
switched to db CollegeDB  
> db["Students"].find()  
[  
  {  
    _id: ObjectId("68c6f1e3d2a7b8c9d0e1f234"),  
    rollNo: 181,  
    name: 'Saqlain',  
    course: 'B.Sc. IT',  
    marks: 90  
  }  
]
```

C) Inserts Records in MongoDB

STEP 1 :- Click “Open MongoDB Shell” at the top-right.

STEP 2 :- Choose the database by entering the command:

use CollegeDB

The **CollegeDB** prompt will start.

STEP 3 :- Use `insertOne()` to insert **one document**.

STEP 4 :- After entering the command, you will get an **acknowledged** response.

```
> db.students.insertOne({})  
< {  
  acknowledged: true,  
  insertedId: ObjectId('6aae2ad4617e6ccf042f0c84')  
}
```

STEP 5 :- Use `insertMany()` to insert **multiple documents**.

STEP 6 :- After entering the command, you will get an **acknowledged** response.

```
> db.students.insertMany([{"rollno":102, name : "Abhi", course:"IT"}, {"rollno":103, name : "Saqlain", course:"IT"}])  
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6aae2c65617e6ccf042f0c85'),  
    '1': ObjectId('6aae2c65617e6ccf042f0c86')  
  }  
}
```

D) View records from the collection

CODE :-

```
db.students.find()
```

```
db.students.updateOne({rollNo:101}, {$set:{course:"CS"}})
```

```
db.students.find()
```

OUTPUT :-

```
db.students.find()
{
  _id: ObjectId('6aae2ad4617e6ccf042f0c84')
}
{
  _id: ObjectId('6aae2c65617e6ccf042f0c85'),
  rollno: 102,
  name: 'Abhi',
  course: 'IT'
}
{
  _id: ObjectId('6aae2c65617e6ccf042f0c86'),
  rollno: 103,
  name: 'Aashu',
  course: 'IT'
}
{
  _id: ObjectId('6aae2e13617e6ccf042f0c87'),
  rollNo: 101,
  name: 'Radha',
  course: 'CS'
}
```

```
db.students.find({rollNo:101})
{
  _id: ObjectId('6aae2e13617e6ccf042f0c87'),
  rollNo: 101,
  name: 'Radha',
  course: 'IT'
}
```

E) Update and Delete Record

CODE :-

```
// Delete student with roll number 104
db.students.deleteOne({rollNo:104})
// Delete one document
db.students.deleteOne({rollNo:103})
// Delete all matching documents
db.students.deleteMany({course:"IT"})
// Update Radha's course from IT to CS
db.students.updateOne({name:"Radha"}, {$set:{course:"CS"}})
```

OUTPUT :-

```
{
  _id: ObjectId('6aae2e13617e6ccf042f0c87'),
  rollNo: 101,
  name: 'Radha',
  course: 'CS'
}
```

F) Connect MongoDB with Node.js

CODE :-

```
const mongoose = require("mongoose");
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
```

OUTPUT :-

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact\pract 7> node server.js
MongoDB Connected Successfully
[
  { _id: new ObjectId('6aae2ad4617e6ccf042f0c84') },
  {
    _id: new ObjectId('6aae2c65617e6ccf042f0c85'),
    rollno: 102,
    name: 'Abhi',
    course: 'IT'
  },
  {
    _id: new ObjectId('6aae2c65617e6ccf042f0c86'),
    rollno: 103,
    name: 'Aashu',
    course: 'IT'
  },
  {
    _id: new ObjectId('6aae2e13617e6ccf042f0c87'),
    rollNo: 101,
    name: 'Radha',
    course: 'CS'
  }
]
```